37 The charge carriers in a metal wire are free electrons.

Which statement about the charge of each free electron is correct?

A The magnitude of the charge increases with the potential difference across the wire.

B The magnitude of the charge is zero unless there is a potential difference across the wire.

C The sign and magnitude of the charge do **not** depend on the potential difference across the wire.

D The sign of the charge depends on the potential difference across the wire.